



CHAPTER 6

Conclusion



The glow of summer has faded, now... and the moonlight jellies carry on towards
the great unknown.

— *Stardew Valley* —



THIS THESIS HAS PRIMARILY concerned itself with improving an established stellar modelling code for future population synthesis projects. In doing so, we discussed several key aspects of binary evolution, which shall be recapped here. To wit, in Chapter 1 we introduced three major themes guiding this research:

- I. Can we study contact binary systems, and refine our treatment of RLOF to overcome numerical issues inherent in these models?
- II. Can we develop a testing regime for the natal supernova kick, which relies on more observations than just isolated pulsar velocities?

In this chapter, we examine these themes in detail and review the implications of them in terms of the results we have obtained.

6.1 | Summary

*Oh! I have slipped the surly bonds of Earth
And danced the skies on laughter-silvered wings;
...
And, while with silent lifting mind I've trod
The high untrespassed sanctity of space,
Put out my hand, and touched the face of God.
— "HIGH FLIGHT", BY JOHN MAGEE JR.*

In Chapter 3, we introduced the notion of helium-rich mass transfer, that being when the mass transfer within a binary continues for a high rate for a long time, thus stripping material from the exposed core of a donor. We found that this produced small differences in the contact phase which were magnified in the post-contact phase. We found that, on net, this produced hotter and at times more luminous accretors in binaries, and concluded that these are potential formation channels for stars such as WN/O3 stars, and may in part explain the discrepancy between spectroscopic and evolutionary masses that persist throughout various modelling efforts.

In Chapter 4, we examined the form and structure of the natal supernova kick. We introduced the [Bray & Eldridge \(2016\)](#) kick and, by direct comparison with the [Hobbs et al. \(2005\)](#) kick, developed a testing pipeline with multiple orthogonal constraints. In turn, we were able to constrain the Bray kick based on four fundamental phenomena: the detected LVK Collaboration merger rate (Section 4.2.1), the observed period-eccentricity distribution of Galactic BNSs (Section 4.2.2), the observed isolated pulsar velocity distribution (Section 4.2.3), and the existence of USSNe (Section 4.2.4). We found that the region of the parameter space that was able to replicate all four phenomena was 1 per cent of the search region, and concluded that the

optimal form of the kick was

$$v_{\text{kick}} = 115_{-55}^{+40} \left(\frac{m_{\text{ejecta}}}{m_{\text{remnant}}} \right) + 15_{-15}^{+10} \text{km s}^{-1}. \quad (6.1)$$

In Chapter 5, we began to search for BPASS matches to six observed close-contact binaries as compiled in [Abdul-Masih et al. \(2022\)](#). We found 31 matches in the BPASS v2.2 dataset. Using these 31 matches as a starting grid, and allowing for ± 0.1 dex in the period grid, we found 15 matching models in total, for LSS 3074, TU Muscae, and V382 Cygni. Of these, three of the models of TU Muscae were found to match the observed masses and radii, with the others all matching on the donors and most frequently failing to match on the radii of the accretors. This was found to not be of large concern, as the companion radius is the most difficult to constrain parameter of the four for an eclipsing binary.

6.2 | Limitations

Though in reviewing the incidents of my administration I am unconscious of intentional error, I am nevertheless too sensible of my defects not to think it probable that I may have committed many errors.

— GEORGE WASHINGTON

There are three major limitations to our study: that we have assumed circular orbits, that we have focused predominantly on O-type stars, and that we have had to make sometimes seemingly arbitrary parameter choices in order for the code to perform. These are detailed in the following subsections.

6.2.1 | Circular Orbits

A common thread throughout the detailed model tracks in this work is the assumption of a circular orbit. Binary stars are not obliged to be circular, and indeed [Moe & Stefano \(2017\)](#) suggest quite the opposite; that the eccentricity distribution varies with the stellar type. Generalising our results to arbitrary eccentricity is, however, a complex undertaking for potentially little payoff. [Eldridge et al. \(2017\)](#) argues, and [Hurley et al. \(2002\)](#) shows, that the result of a binary computation is the same for an elliptical orbit and a circular orbit, if the circle has a radius equal to the semi-latus rectum of the ellipse. This is illustrated geometrically in Figure 6.1.

6.2.2 | Our parameter choices

Most of our parameter choices have in some way been physically motivated; for instance the choice of k in Section 2.4.3, corresponding to the efficiency of RLOF, was chosen via a combination of work by [Hurley et al.](#)

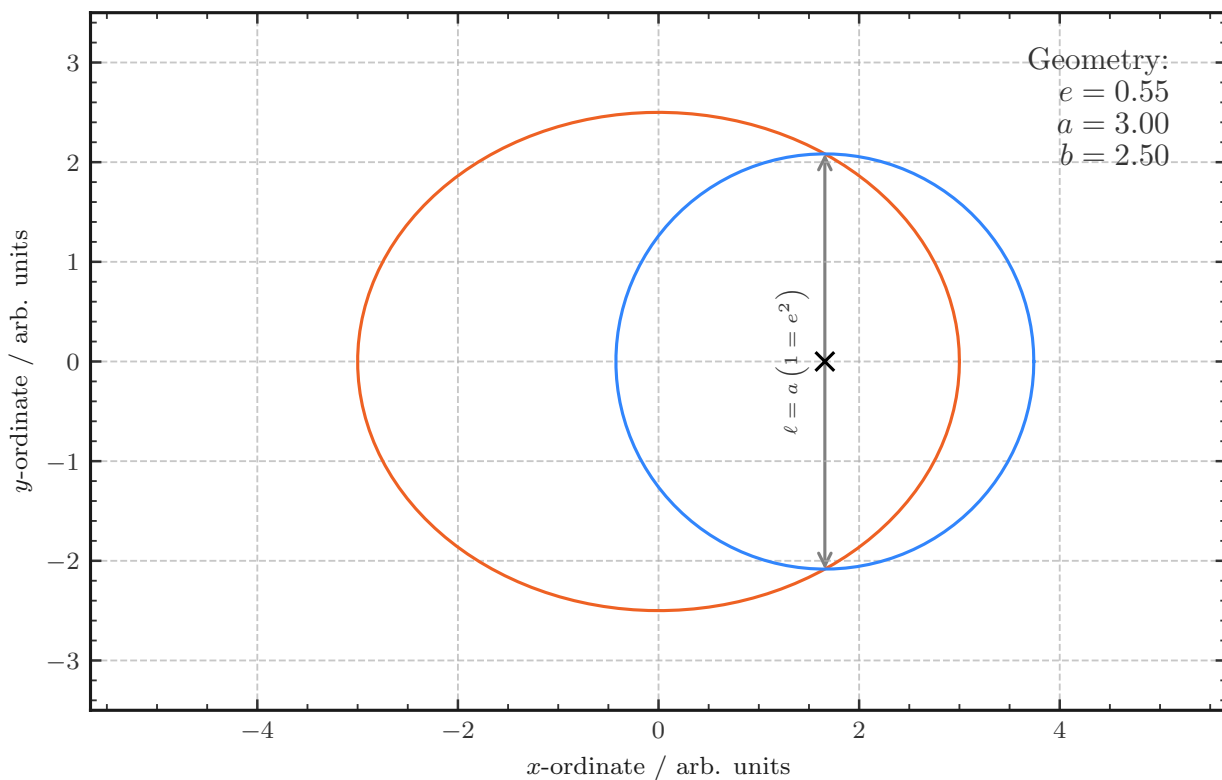


Figure 6.1: The geometry of an ellipse and the circle with a radius equal to the semi-latus rectum. The ellipse was chosen arbitrarily to have $a = 3, b = 2.5$. The black cross is at the focus of the ellipse, which is also the center of the circle.

(2002) and [Claeys et al. \(2014\)](#). Through testing, it has become apparent that the [Kolb & Ritter \(1990\)](#) treatment of mass transfer may be more appropriate; an implementation of this as a mass loss scheme would be a potential avenue for future exploration of the constraints of mass transfer.

Stellar modelling is a nonlinear task, and the clearest example of this in this thesis is the thermohaline mixing coefficients α_{th} in Chapter 3, where the interior composition profile was changed quite substantially depending on the strength of the thermohaline mixing. Most of these somewhat arbitrary parameter choices are a reflection of uncertainty in the physics; thermohaline mixing in particular is a difficult process to constrain.

6.3 | Suggested Future Work

*A still more glorious dawn awaits
– not a sunrise, but a galaxy rise.*

— CARL SAGAN

I can slay the Jabberwocky.

— LAST OF THE SIX IMPOSSIBLE THINGS

There are multiple avenues open to future research questions with the STARS code, and in this section we outline but a few.

Chapter 4 examined the form and structure of the natal supernova kick, with a particular emphasis on [Bray & Eldridge \(2016\)](#) type kicks. Looking forward, LVK Collaboration O4 will soon release, offering constraints on the observed BNS merger rate density. In addition, the same pipeline can be performed on other kick prescriptions that offer free parameters, such as [Verbunt et al. \(2017\)](#), which would allow for their refinement.

Chapter 5 explored a rather limited parameter space. The question thus remains, *to what extent do the conclusions drawn hold true on a population of binary stars*, and to much the same extent, *do the population prevalences derived from the BPASS v2 models match those from a grid of these models*? Exploration of the parameter space is firmly a big data project. For a reasonably trivial grid of $m_1 \in [5, 100] \text{ M}_\odot$ in steps of $5 \text{ M}_\odot$, $q \in [0, 0.9]$ in steps of 0.1, and $\log P \in [0.5, 3]$ in steps of 0.5 dex, one arrives at 1200 models. Models typically take, conservatively, between 20 min and 60 min to run on the cluster, which has up to 72 cores available for use. Table 6.1 shows some estimated runtimes for different grid fidelities given this setup. An investigation into the population properties would likely need to use model B as a coarse grid and model D as a finer grid, though the span of periods is wider here than is strictly necessary.

Model Name	Grid Specification				Runtime (DD:HH:MM:SS)		
	Δm_1	Δq	$\Delta \log P$	N_{models}	20 min	40 min	60 min
A	5	0.1	0.5	1200	00:05:33:20	00:11:06:40	00:16:40:00
B	5	0.1	0.1	5200	01:00:04:26	02:00:08:53	03:00:13:20
C	1	0.1	0.5	5760	01:02:40:00	02:05:20:00	03:08:00:00
D	1	0.1	0.1	24960	04:19:33:20	09:15:06:40	14:10:40:00

Table 6.1: The runtimes for a grid of STARS models on a 72-core cluster, assuming each model takes 20 min/40 min/60 min respectively. This calculation assumes all cores begin operating simultaneously, and that runs are homogeneous; that any two runs have the same run time. In this grid, $m_1 \in [5, 100] \text{ M}_\odot$, $q \in [0, 0.9]$, and $\log P \in [0.5, 3]$.

Looking towards the eastern horizon, the BPASS project and other similar population synthesis codes will soon be releasing new versions. The STARS code is, compared to MESA, more efficient on a single core as it is simpler than MESA. It also boasts the wider BPASS code suite as an extension of this, with the associated population synthesis and spectral synthesis codes. The STARS code thus seems uniquely positioned to offer new and exciting insights into the field of binary stellar evolution.

And that is something you might not have known.

— TOM SCOTT

Appendices

